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Github link: https://github.com/RMIT-Vietnam-Teaching/assignment-1-individual-quoc-huy2004

Title

Application Description

1. **Purpose**:

    - The primary purpose of the BidBlitz application is to provide a simple auction management system that allow users to buy and sell items through a competitive bidding process. The system enables sellers to list items for auction and allows buyers to place bids, with the highest bidder winning when the auction ends.
    - Many individuals and sellers do not have an easy-to-use platform to manage auctions and tracking bidding activities. While traditional manual processes are inefficient and lack of transparency, the application is created to solve:
        - Creating an auto auction lifecycle (creation, bidding, and closing).
        - Ensuring fair competition through bid validation.
        - Tracking users, items, auctions, and payments in a structured way.
        - Reducing manual effort and improving accuracy.
    - The system provides the following core features:
        - User management (View users, Top-up balance).
        - Item creation and categorization.
        - Auction creation and management.
        - Bid placement with validation rules.
        - Automatic processing of auctions based on time.
        - Payment generation for completed auctions.
        - Reporting and data tracking.
2. **Key Features**:

    - The BidBlitz application includes several core features that support management system and ensure smooth system operation:
        - **User Input Handling:**
            - Interactive menu-driven interface using **Scanner**.
            - Allow users to select menu functions (e.g., place bid, create item) and enter information such as: IDs, prices, and other required data
            - Having basic validation (e.g., invalid choice, incorrect input format, real money logic).

- **File Operations:**
  - Read initial data from text files (e.g., users.txt, items.txt, auctions.txt).
  - Saves updated data back to files upon program exit.
  - Ensures data persistence during running program.
- **Auction Management:**
  - Create and manage auctions for items.
  - Track auction status (**ACTIVE, SOLD, UNSOLD**).
  - Automatically process auctions when end time is called.
- **Bidding System:**
  - Users can place bids on active auctions.
  - Validates bids:
    1. Must be higher than current highest bid.
    2. Cannot bid on own items.
  - Tracks highest bid and leading bidders.
- **Payment Processing:**
  - Automatically generates payment records when an auction is sold.
  - Calculates:
    1. Total price.
    2. Commission fee based on category.
    3. Seller payout.
  - Track payment status (e.g., PENDING, COMPLETE).
- **Item and Category Management:**
  - Create new items with details such as title, price, and category.
  - Organize items by categories.
  - Associate items with sellers.
- **Reporting and Summary:**
  - Displays system statistics:
    1. Total users.
    2. Total items.
    3. Total auctions.
- **Use of OOP Principles:**
  - Encapsulation through classes (User, Item, Auction, etc.)
  - Separation of responsibilities (e.g., AuctionManagerImpl, DataLoader).
  - Reusable and modular design.

3. **Target Audience**:

   - The BidBlitz application is intended for individuals and small-scale sellers who want to manage auctions in a simple and efficient way. It is also suitable for buyers who are interested in participating in bidding processes to purchase items.

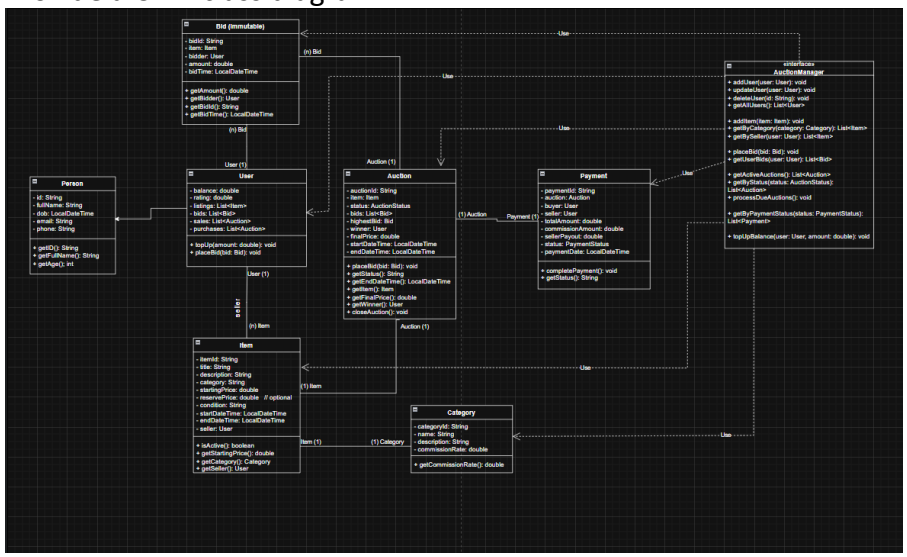
- Additionally, the system is designed for administrators or system operators who need to manage users, monitor auctions, and track payments within a structured place.

4. **Scope**: Outline the scope of the project, including any constraints or limitations.

   - BidBlitz application is created with the scope of covering the core functionalities required to manage a basic auction system in a console-based environment. The system allows administrators and users to create items, manage auctions, place bids, process auction results, and generate payment records.
   - Constraints and Limitations:
     - The system is console-based (No graphical user interface).
     - Data is stored in text files instead of a database.
     - Limited input validation and error handling.
     - No real-time bidding or concurrent user support
     - Auction timing depends on system time (not real-time server-based).
     - Security features (e.g., authentication, authorization) are not implemented.

Application Diagram

1. Provide a UML class diagram



2. Activity/ Use-case Diagram

API List (With Brief Description)

1. **Method Overview**:

API/Method Name	Purpose	Input	Output
getAllUsers()	Retrieve all users in the system	None	List<User>
topUpBalance(User, double)	Add funds to a user's balance	User, double	void
addUser(User)	Add a new user to the system	User	void
getAllItems()	Retrieve all items	None	List<Item>
addItem(Item)	Add a new item	Item	void
getByCategory(Category)	Get items by category	Category	List<Item>
getBySeller(User)	Get items listed by a seller	User	List<Item>
placeBid(Bid)	Place a bid on an auction	Bid	void
getUserBids(User)	Retrieve all bids created by a user	User	List<Bid>
getActiveAuctions()	Get all active auctions	None	List<Auction>
getAllAuctions()	Retrieve all auctions	None	List<Auction>
addAuction(Auction)	Create a new auction	Auction	void
processDueAuctions()	Process auctions whose end time has passed	None	void

createPayment(Auction)	Generate payment after auction is sold	Auction	void
getAllPayments()	Retrive all payments	None	List<Payment>
getByPaymentStatus(String)	Filter payments by status	status	List<Payment>
closeAuction()	Close auction and determine result	None	void
placeBid(Bid) *(Auction class)*	Add bid and update highest bid	Bid	void
completePayment()	Mark payment as completed	None	void
loadAll(AuctionManagerImpl)	Load all data to files	Manager	void
saveAll(AuctionManagerImpl)	Save all data to files	Manager	void
findUser(AuctionManagerImpl, String)	Find user by ID	Manager, userID	User
findItem(AuctionManagerImpl, String)	Find item by display ID	Manager, itemID	Item

- **External Libraries**:
  No third-party libraries are used in this application because the system is implemented using only standard Java libraries to handle data structures, file operations, and date-time processing.

Application Output (Screenshots)

1. **Screenshots**:
   - Main Menu

     - Shows the menu options (1-9) and an exit option (0)7.

- Demonstrates user input handling.

```
===== BIDBLITZ SYSTEM =====
1. Show all users
2. Show all active auctions
3. Place a bid
4. Process auctions
5. Top up user balance
6. Create item
7. Create auction
8. View Payments
9. Reports
0. Exit
Choose:
```

- Display Users:
  - Output of all users with balances.

  - Demonstrates data loading and retrieval.

```
Choose: 1
U1 - Nguyen Van A - Balance: 1500.0
U2 - Tran Thi B - Balance: 1500.0
U3 - Le Van C - Balance: 2000.0
U4 - Pham Thi D - Balance: 1200.0
U5 - Hoang Van E - Balance: 1800.0
U6 - Nguyen Thi F - Balance: 2200.0
U7 - Tran Van G - Balance: 900.0
U8 - Le Thi H - Balance: 1300.0
U9 - Pham Van I - Balance: 1700.0
U10 - Hoang Thi K - Balance: 2500.0
U11 - Nguyen Van L - Balance: 3000.0
U12 - Tran Thi M - Balance: 1100.0
U13 - Le Van N - Balance: 1600.0
U14 - Pham Thi O - Balance: 1400.0
U15 - Hoang Van P - Balance: 2100.0
```

- Display all active auctions:
- Show all available auctions along with the current status.

```
Choose: 2
=== ACTIVE AUCTION ===
I1 | Laptop | $500.0 | No bids | ACTIVE | Ends: 2026-04-04T17:25:24.170714100
I3 | Shirt | $50.0 | No bids | ACTIVE | Ends: 2026-04-04T17:25:24.198631700
I4 | Table | $150.0 | No bids | ACTIVE | Ends: 2026-04-04T17:25:24.198631700
I5 | Toy Car | $30.0 | No bids | ACTIVE | Ends: 2026-04-04T17:25:24.198631700
I6 | Basketball | $70.0 | No bids | ACTIVE | Ends: 2026-04-04T17:25:24.198631700
I7 | Lipstick | $25.0 | No bids | ACTIVE | Ends: 2026-04-04T17:25:24.198631700
I8 | Car Tire | $200.0 | No bids | ACTIVE | Ends: 2026-04-04T17:25:24.198631700
I9 | Guitar | $400.0 | No bids | ACTIVE | Ends: 2026-04-04T17:25:24.198631700
I10 | Notebook | $10.0 | No bids | ACTIVE | Ends: 2026-04-04T17:25:24.200138100
I11 | Dog Food | $40.0 | No bids | ACTIVE | Ends: 2026-04-04T17:25:24.200138100
I12 | Game CD | $60.0 | No bids | ACTIVE | Ends: 2026-04-04T17:25:24.200138100
I13 | Vitamins | $20.0 | No bids | ACTIVE | Ends: 2026-04-04T17:25:24.200138100
I14 | Snacks | $15.0 | No bids | ACTIVE | Ends: 2026-04-04T17:25:24.200138100
I15 | Suitcase | $120.0 | No bids | ACTIVE | Ends: 2026-04-04T17:25:24.200138100
```

- Place a bid (Successful case):
  - Show:
    - Item list.
    - User input.
    - Success message.

➢ Highest bid update.

```
Choose: 3
=== ITEMS ===
I1 - Laptop - Starting Price: $500.0
I2 - Phone - Starting Price: $300.0
I3 - Shirt - Starting Price: $50.0
I4 - Table - Starting Price: $150.0
I5 - Toy Car - Starting Price: $30.0
I6 - Basketball - Starting Price: $70.0
I7 - Lipstick - Starting Price: $25.0
I8 - Car Tire - Starting Price: $200.0
I9 - Guitar - Starting Price: $400.0
I10 - Notebook - Starting Price: $10.0
I11 - Dog Food - Starting Price: $40.0
I12 - Game CD - Starting Price: $60.0
I13 - Vitamins - Starting Price: $20.0
I14 - Snacks - Starting Price: $15.0
I15 - Suitcase - Starting Price: $120.0
Enter item ID: I2
Enter user ID: U1
Enter bid amount: 500
Bid placed successfully!
New highest bid: $500.0
Leading bidder: U1
```

- Check the auctions status (Right after place a bid):
- Update the leading bidder and new price.

```
Choose: 2
=== ACTIVE AUCTION ===
I1 | Laptop | $500.0 | No bids | ACTIVE | Ends: 2026-04-04T17:39:03.967442200
I2 | Phone | $500.0 | U1 | ACTIVE | Ends: 2026-04-04T17:39:03.997307400
I3 | Shirt | $50.0 | No bids | ACTIVE | Ends: 2026-04-04T17:39:03.997307400
I4 | Table | $150.0 | No bids | ACTIVE | Ends: 2026-04-04T17:39:03.997307400
I5 | Toy Car | $30.0 | No bids | ACTIVE | Ends: 2026-04-04T17:39:03.997307400
I6 | Basketball | $70.0 | No bids | ACTIVE | Ends: 2026-04-04T17:39:03.997307400
I7 | Lipstick | $25.0 | No bids | ACTIVE | Ends: 2026-04-04T17:39:03.997307400
I8 | Car Tire | $200.0 | No bids | ACTIVE | Ends: 2026-04-04T17:39:03.997307400
I9 | Guitar | $400.0 | No bids | ACTIVE | Ends: 2026-04-04T17:39:03.997307400
I10 | Notebook | $10.0 | No bids | ACTIVE | Ends: 2026-04-04T17:39:03.997307400
I11 | Dog Food | $40.0 | No bids | ACTIVE | Ends: 2026-04-04T17:39:03.997307400
I12 | Game CD | $60.0 | No bids | ACTIVE | Ends: 2026-04-04T17:39:03.997307400
I13 | Vitamins | $20.0 | No bids | ACTIVE | Ends: 2026-04-04T17:39:03.997307400
I14 | Snacks | $15.0 | No bids | ACTIVE | Ends: 2026-04-04T17:39:03.998799
I15 | Suitcase | $120.0 | No bids | ACTIVE | Ends: 2026-04-04T17:39:03.998799
```

-

- Error Handling:
- Bid too low.
- User and Item not found.

```
=== ITEMS ===
I1 - Laptop - Starting Price: $500.0
I2 - Phone - Starting Price: $300.0
I3 - Shirt - Starting Price: $50.0
I4 - Table - Starting Price: $150.0
I5 - Toy Car - Starting Price: $30.0
I6 - Basketball - Starting Price: $70.0
I7 - Lipstick - Starting Price: $25.0
I8 - Car Tire - Starting Price: $200.0
I9 - Guitar - Starting Price: $400.0
I10 - Notebook - Starting Price: $10.0
I11 - Dog Food - Starting Price: $40.0
I12 - Game CD - Starting Price: $60.0
I13 - Vitamins - Starting Price: $20.0
I14 - Snacks - Starting Price: $15.0
I15 - Suitcase - Starting Price: $120.0
Enter item ID: I16
Error: Item not found
```

```
=== ITEMS ===
I1 - Laptop - Starting Price: $500.0
I2 - Phone - Starting Price: $300.0
I3 - Shirt - Starting Price: $50.0
I4 - Table - Starting Price: $150.0
I5 - Toy Car - Starting Price: $30.0
I6 - Basketball - Starting Price: $70.0
I7 - Lipstick - Starting Price: $25.0
I8 - Car Tire - Starting Price: $200.0
I9 - Guitar - Starting Price: $400.0
I10 - Notebook - Starting Price: $10.0
I11 - Dog Food - Starting Price: $40.0
I12 - Game CD - Starting Price: $60.0
I13 - Vitamins - Starting Price: $20.0
I14 - Snacks - Starting Price: $15.0
I15 - Suitcase - Starting Price: $120.0
Enter item ID: I5
Enter user ID: U20
Enter bid amount: 80
Error: Bidder cannot be null
```

```
=== ITEMS ===
I1 - Laptop - Starting Price: $500.0
I2 - Phone - Starting Price: $300.0
I3 - Shirt - Starting Price: $50.0
I4 - Table - Starting Price: $150.0
I5 - Toy Car - Starting Price: $30.0
I6 - Basketball - Starting Price: $70.0
I7 - Lipstick - Starting Price: $25.0
I8 - Car Tire - Starting Price: $200.0
I9 - Guitar - Starting Price: $400.0
I10 - Notebook - Starting Price: $10.0
I11 - Dog Food - Starting Price: $40.0
I12 - Game CD - Starting Price: $60.0
I13 - Vitamins - Starting Price: $20.0
I14 - Snacks - Starting Price: $15.0
I15 - Suitcase - Starting Price: $120.0
Enter item ID: I9
Enter user ID: U1
Enter bid amount: 200
Error: Bid too low
```

- Process Auctions:

- Show auction result:
  - SOLD.
  - Winner + price.

```
Choose: 4
I2 is sold to U1 for $500.0
=== PROCESSED AUCTIONS ===
I1 | Laptop | ACTIVE
I2 | Phone | SOLD
I3 | Shirt | ACTIVE
I4 | Table | ACTIVE
I5 | Toy Car | ACTIVE
I6 | Basketball | ACTIVE
I7 | Lipstick | ACTIVE
I8 | Car Tire | ACTIVE
I9 | Guitar | ACTIVE
I10 | Notebook | ACTIVE
I11 | Dog Food | ACTIVE
I12 | Game CD | ACTIVE
I13 | Vitamins | ACTIVE
I14 | Snacks | ACTIVE
I15 | Suitcase | ACTIVE
```

- View Payments:
- Demonstrates payment calculation logic.

```
Choose: 8
Item: I2 | Buyer: U1 | Seller: U2 | Total: $500.00 | Commission: 5% ($25.00) | Status: COMPLETED
```

- Create Item/Auctions: (number 6 and 7)

```
=== CREATE ITEM ===
Title (e.g: Laptop, Phone): Stuff
Starting price (e.g: 100, 200.5): 199
=== USERS ===
U1 - Nguyen Van A
U2 - Tran Thi B
U3 - Le Van C
U4 - Pham Thi D
U5 - Hoang Van E
U6 - Nguyen Thi F
U7 - Tran Van G
U8 - Le Thi H
U9 - Pham Van I
U10 - Hoang Thi K
U11 - Nguyen Van L
U12 - Tran Thi M
U13 - Le Van N
U14 - Pham Thi O
U15 - Hoang Van P
Enter seller ID (e.g: U1, U2): U3
=== CATEGORIES ===
C1 - Electronics
C2 - Fashion
C3 - Books
C4 - Furniture
C5 - Toys
C6 - Sports
C7 - Beauty
C8 - Automotive
C9 - Music
C10 - Office
C11 - Pet
C12 - Gaming
C13 - Health
C14 - Food
C15 - Travel
Enter category ID (e.g: C1, C2):  C4
 Item created successfully!
I16 - Stuff
```

```
Choose: 7
Enter item ID: I16
Auction created!
```

- Show report:

```
Choose: 9
=== REPORT ===
Total users: 15
Total items: 16
Total auctions: 16
```

- Additionally, users can top-up to their account:

```
Choose: 5
Enter user ID: U1
Enter amount: 400
Balance updated!
```

```
Choose: 1
U1 - Nguyen Van A - Balance: 1900.0
```

Any Drawback and Future Work

1. **Drawbacks**:
   - Console-Based Interface:
     1. The application uses a text-based interface, which is less user-friendly compared to modern graphical or web interfaces.
        - This limitation exists because the project focuses on core logic implementation rather than UI design.
   - File-Based Data Storage:
     1. Data is store in text files instead of a database.
     2. This can lead to inefficiency, lack of scalability, and potential data inconsistency.
        - The limitation exists due to simplicity and the scope of the assignment.
   - Limited Validation and Error Handling:
     1. Input validation is basic and may not handle all cases.
        - This is due to time constraints and prioritizing core functionality.
   - No Real-Time or Concurrent Bidding:
     1. The system does not support multiple users bidding at the same time.
     2. Auctions are processed manually rather than in real-time.
        - This limitation exists because concurrency and real-time systems require more advance techniques.
2. **Future Work**:
   - Develop a Web-Based Interface:
     1. Convert the system into a web or mobile application with a graphical user interface.

        - This would improve usability and accessibility for a wider range of users.

- Integrate a Database System:
    1. Replace text file storage with a database (e.g., MySQL or MongoDB).
        - This would improve data reliability, scalability, and performance.
- Implement Real-Time Auction Features:
    1. Add real-time bidding with automatic updates and countdown timers.
        - This would make the system more realistic.